

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Anti-seize compound, Part Number 3824879, or equivalent

Additional Service Items

- No additional service items required.

General Information

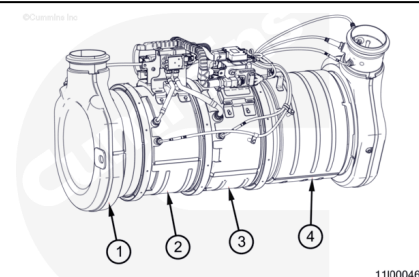
Due to the number of exhaust aftertreatment applications, this procedure has been written to be generic. **Not** all illustrations within this document will represent the application being serviced.

The aftertreatment exhaust assembly is composed of four sections. These sections are:

1. The aftertreatment diesel oxidation catalyst (DOC)
2. The aftertreatment diesel particulate filter (DPF)
3. The aftertreatment decomposition reactor tube
4. The selective catalytic reduction (SCR).

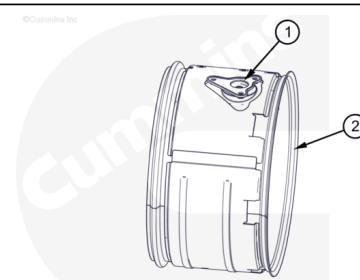
The decomposition tube (3) is the section of exhaust pipe between the DPF (2) and SCR catalyst (4).

The aftertreatment decomposition tube is designed to help atomize and mix the diesel exhaust fluid (DEF) being sprayed into the exhaust stream for complete NO_x conversion.



The aftertreatment decomposition tube is composed of two important areas.

1. The aftertreatment DEF dosing valve mount: The position at which the aftertreatment DEF dosing valve is mounted and allowed to spray DEF into the exhaust stream.

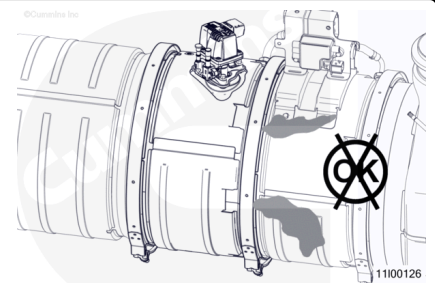


2. The aftertreatment decomposition tube mixer: The mixer creates a swirl pattern exhaust stream to help mix the DEF and engine exhaust.

Different vehicles will require different aftertreatment configurations. Some of the aftertreatment decomposition tubes may have elbows clamped to either end of the tube.

Initial Check

Inspect all the joints for any signs of leaks. Leaking DEF will leave white deposits. If a DEF leak is found, see the Clean and Inspect for Reuse section of this procedure.



Preparatory Steps

⚠ WARNING ⚠

During regeneration, exhaust gas temperature could reach 800°C [1500°F], and exhaust system surface temperature could exceed 700°C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle has stopped moving. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

The aftertreatment DEF dosing valve is a fragile item. Use great care when handling this part. Avoid putting stresses on the fluid connectors as damage to the component may result.

Note : It is acceptable to remove the aftertreatment dosing valve without removing the DEF lines. However, support the aftertreatment DEF dosing valve. Do not let it suspend freely. It can damage the fluid connectors and lines.

Allow for DEF to recirculate and depressurize after engine shut down and prior to DEF system servicing. This can take up to 15 minutes.

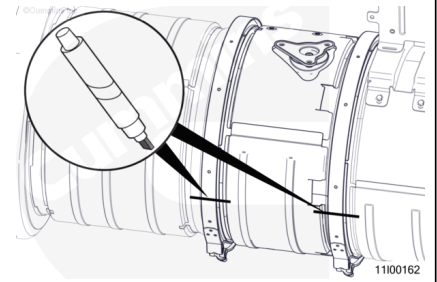
There is **always** a chance of residual pressure being present. Open fittings slowly to allow any pressure to bleed off before removing any connections

- Disconnect the batteries. See equipment manufacturer service information.
- Clean the area with compressed air. Remove any debris from the aftertreatment decomposition tube.
- Remove the pressure and return DEF lines from the DEF dosing valve. See equipment manufacturer service information.
- Remove the aftertreatment DEF dosing valve from the aftertreatment decomposition tube. Refer to Procedure 011-059 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-059.html>)
- Remove the outlet DPF temperature sensor probe from the decomposition tube. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-449.html>)
- Remove the outlet differential pressure sensor tube nut from the decomposition tube. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-047.html>)
- Remove any wire ties or clamps that secure the sensor wires to the decomposition tube.
- Disconnect the interface harness from the outlet NOx sensor. Refer to Procedure 019-477 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-477.html>)
- Disconnect the interface harness from the particulate matter sensor. Refer to Procedure 019-477 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-477.html>)
- Remove the capscrews that secure the sensor mounting bracket to the decomposition tube. Refer to Procedure 011-142 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-142.html>)

Remove

⚠ CAUTION ⚠

Do not use an air tool to remove the V-band clamp nut. An air tool will damage the threads. Apply thread lubricant to the V-band clamp threads prior to nut removal.



Note : Before removing the aftertreatment decomposition tube note the location and orientation of the aftertreatment decomposition tube and related components. Apply identifying markings to aid in installation.

Note : Depending on the exhaust configuration, it may be necessary to loosen exhaust hangers or remove additional elbows and exhaust piping securing the rest of the aftertreatment before removing the aftertreatment decomposition tube.

Remove the exhaust clamps from both ends of the aftertreatment decomposition tube.

Remove the aftertreatment decomposition tube.

Discard the gaskets.

Clean and Inspect for Reuse

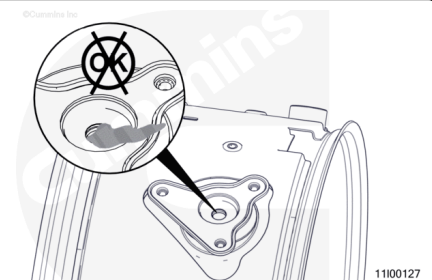
Note : If troubleshooting a fault code that has a step that indicates to check the aftertreatment decomposition for DEF deposits, clean the aftertreatment decomposition tube of DEF deposits, regardless of the inspection criteria in the steps below.

Inspect the aftertreatment DEF dosing valve spray port in the aftertreatment decomposition tube for DEF deposits.

If there are DEF deposits in the aftertreatment DEF dosing valve spray port, clean the aftertreatment decomposition tube according to the cleaning steps below.

If the aftertreatment DEF dosing valve spray port is blocked with DEF deposits, there may be issues with the dosing valve. Refer to Procedure 011-059 in Section 11.

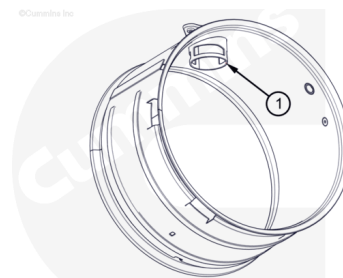
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Inspect the aftertreatment DEF dosing valve spray chamber (1) in the aftertreatment decomposition tube. If there is a DEF deposit protruding into the main decomposition tube area that has a free flow path for pressurized DEF stream and there are no active fault codes, no action is required.

Note : Use a light source when looking into the aftertreatment decomposition tube to inspect for a free flow path for a pressurized DEF stream.

If there is a DEF deposit protruding into the main decomposition tube area restricting the free flow path for a pressurized DEF stream, clean the aftertreatment decomposition tube according to the cleaning steps below.

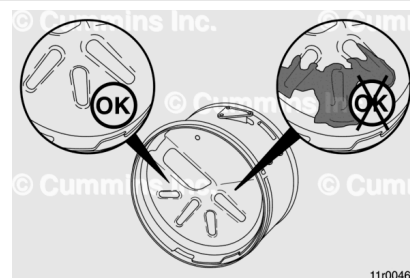


11100128

Inspect the aftertreatment decomposition tube main area around the mixer. If the mixer area is less than 50 percent blocked by DEF deposits, no action is necessary.

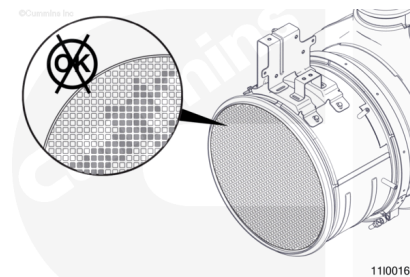
If the DEF deposits in the mixer area are blocking more than 50 percent of the aftertreatment decomposition tube, regardless if any related active fault codes are present, the aftertreatment decomposition tube **must** be cleaned. See the cleaning steps below.

If the DEF deposits in the mixer area are blocking more than 50 percent of the aftertreatment decomposition tube, there may be issues with the dosing valve. Refer to Procedure 011-059 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-059.html>)



11100465

Inspect the intake of the aftertreatment SCR catalyst for DEF deposits. Refer to Procedure 011-036 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-036.html>)

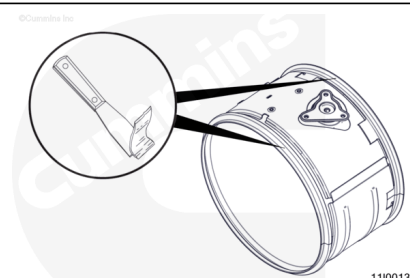


11100163

⚠ WARNING ⚠

Wear safety glasses or a face shield, as well as protective clothing, to prevent personal injury when using a steam cleaner or high-pressure water.

⚠ CAUTION ⚠



11100131

Do not use a metallic object to clean the aftertreatment decomposition tube. This will scratch the surface of the aftertreatment decomposition tube which may cause future excessive DEF deposits.

If DEF deposits are present, carefully scrape with a non-metallic object to remove the majority of the DEF deposits.

Use a pressure washer to dissolve the remaining DEF deposits.

Note : It is acceptable to have a small amount of DEF deposits remaining in the mixer area after the cleaning steps. The remaining DEF deposits can be removed by performing a stationary regeneration. Reference the Finishing Steps below.

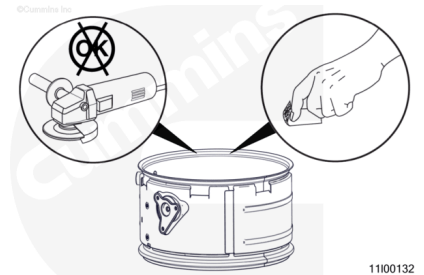
Use **only** water to remove deposits.

Do **not** submerge the aftertreatment decomposition tube in water or solvents, which can saturate the insulation.

Inspect the exhaust flanges for corrosion and other damage.

Use an abrasive pad to remove any residual gasket material from the flanges on the aftertreatment decomposition tube.

Do **not** grind on the flange surface. This can damage the flange and cause the connection to leak.

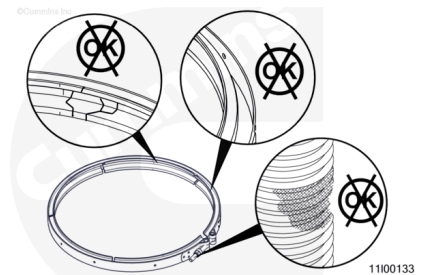


Inspect the V-band clamps and mounting straps for signs of overextension.

The band **must not** be bent or damaged.

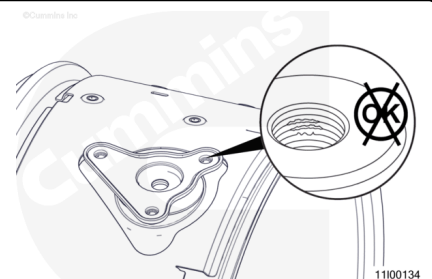
Inspect the V-band clamp and mounting strap threads for damage.

Replace the V-band clamp or mounting strap if damage is found.



⚠ CAUTION ⚠

Do not use a metallic object to clean the aftertreatment decomposition tube dosing valve mounting area. Damage to the gasket/isolator sealing surface area can result.



Inspect the aftertreatment DEF dosing valve mounting area.

Inspect for old gasket/isolator material and corrosion.

Clean with an abrasive pad.

Check the threaded capscrew holes for damaged threads.

If the capscrew holes are damaged, the aftertreatment decomposition tube can be repaired.

Inner coil threads, M6 X 1, can be used to repair any damaged threads.

Use **only** the wire type coil inserts because the amount of material around the threaded capscrew hole is limited.

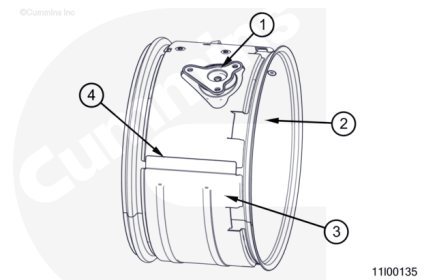
Follow manufacturer recommended guidelines to install the coiled threaded insert.

Note : The aftertreatment decomposition tube should be smooth and free from scratches and other damage.

If the aftertreatment decomposition tube has **any** structural defects or material degradation, replace the aftertreatment decomposition tube.

Check the following areas:

1. Aftertreatment dosing valve mount
2. Mixer
3. Insulation
4. Welded joints.



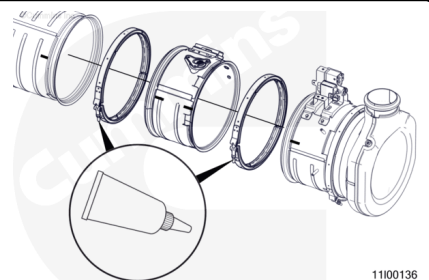
Install

⚠ CAUTION ⚠

Do not use an air tool to tighten the V-band clamp. An air tool will damage the threads.

Install the aftertreatment decomposition tube with new gaskets.

If other parts of the aftertreatment were adjusted in the removal process of the aftertreatment decomposition tube, slide them back into position.



Align the aftertreatment decomposition tube using the alignment marks created in the previous steps. **Only** use the alignment arrows if the DPF was **not** moved or rotated during the removal of the decomposition tube.

Apply a coat of anti-seize compound on the threads of the V-band clamps.

Install the clamps.

Torque Value: 20 n•m [177 in-lb]

Note : Do not place the T-bolt section of the V-band clamp near the wiring harness.

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the cap screws that secure the sensor mounting bracket to the decomposition tube. Refer to Procedure 011-142 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-142.html>)
- Connect the interface harness to the outlet NOx sensor. Refer to Procedure 019-477 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-477.html>)
- Connect the interface harness to the particulate matter sensor. Refer to Procedure 019-477 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-477.html>)
- Install the outlet DPF temperature sensor probe to the decomposition tube. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-449.html>)
- Install the outlet differential pressure sensor tube nut to the decomposition tube. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-047.html>)
- Install any wire ties or capscrews that secure the sensor wires to the decomposition tube.
- Tighten any aftertreatment hangers that could possibly have been loosened in the removal process. See equipment manufacturer service information.
- Install the aftertreatment DEF dosing valve. Refer to Procedure 011-059 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-059.html>)
- Install the pressure and return DEF lines to the DEF dosing valve. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Start the engine. Check for exhaust leaks.

Note : If DEF deposits were present, and cleaning of the aftertreatment decomposition tube was required, complete a stationary regeneration to make sure that any of the remaining DEF deposit has been removed from the aftertreatment system. Continue to monitor for leaks at the exhaust joints.

Last Modified: 10-Oct-2019
